network coverage. The hotel offers valet parking, but you don't trust the attendant, so you only want to allow them limited ability to drive the expensive car you hired. Because your access token is a macaroon, you can simply append caveats to it restricting the token to expire in 10 minutes and only allow the car to be driven in a quarter-mile radius of the hotel.

Macaroons are a great solution for offline authorization because caveats can be added by devices at any time without any coordination and can then be locally verified by devices without needing to contact a central service. Third-party caveats can also work well in an IoT application, because they require the client to obtain proof of authorization from the 3rd-party API. This authorization can be obtained ahead of time by the client and then verified by the device by checking the discharge macaroon, without needing to directly contact the 3rd party.

13.5 Summary

- Devices can be identified using credentials associated with a device profile. These credentials could be an encrypted pre-shared key or a certificate containing a public key for the device.
- Device authentication can be done at the transport layer, using facilities in TLS, DTLS, or other secure protocols. If there is no end-to-end secure connection, then you'll need to implement your own authentication protocol.
- End-to-end device authentication must ensure freshness to prevent replay attacks. Freshness can be